CFA® LEVEL III · PORTFOLIO MANAGEMENT PATHWAY · FOR THE FEBRUARY 2027 EXAM

CFA L3 Portfolio Management Pathway Summary

Every reading (P1–P8) of the Schweser 2025 Portfolio Management pathway notes, explained from zero — 78 subtopics. The pathway is 30–35% of the Level III exam (about 7 of the 20 scored question sets). Each subtopic opens with 'In plain words', then tables, formulas, worked examples and exam traps. Stars combine exam weight, essay likelihood and depth in the curriculum.

★★★★★ must know, asked often ★★★★☆ high ★★★☆☆ medium ★★☆☆☆ low ★☆☆☆☆ skim

‡ rule or number that changes often; check workbook edition

Ch	Chapter	Marks	Priority	Subtopics	5★
0	Exam Trend & How to Use This Summary	—	★★★★★	3	2
1	P1 Index-Based Equity Strategies	4% of PM	★★★☆☆	4	1
2	P2 Active Equity Investing: Strategies	12% of PM	★★★★☆	9	7
3	P3 Active Equity Investing: Portfolio Construction	11% of PM	★★★★☆	11	6
4	P4 Liability-Driven & Index-Based Fixed Income	14% of PM	★★★★★	8	6
5	P5 Yield Curve Strategies	13% of PM	★★★★★	9	8
6	P6 Fixed-Income Active Management: Credit Strategies	19% of PM	★★★★★	12	8
7	P7 Trade Strategy and Execution	12% of PM	★★★★☆	8	7
8	P8 Case Study: Institutional (University Endowment)	15% of PM	★★★★★	10	9
F	Formula Sheet (all readings)	revise last	★★★★★	4	4

5★ list (58) — read these first

Ch 0 · Level III exam structure (Feb 2027) · Where the pathway marks come from

Ch 1 · Full replication, stratified sampling, optimisation

Ch 2 · Fundamental vs quantitative approaches · Bottom-up strategies: value & growth sub-styles · Factor-based active strategies & the hedged portfolio · Activist strategies · Statistical arbitrage & event-driven strategies · Building a quantitative strategy; information coefficient · Style classification: holdings-based vs returns-based

Ch 3 · Sources of active return & the building blocks · Fundamental law of active management · Active Share vs active risk · Risk budgeting: absolute risk · AUM, liquidity, turnover and market impact · Long-only, long/short, long extension, market-neutral

Ch 4 · Immunisation & duration basics · Single-liability immunisation rules & structural risk · Multiple liabilities: cash-flow vs duration matching · Closing a duration gap with futures · Swaps, swaptions and collars for the duration gap · Bond indexing: why it's hard & how to match

Ch 5 · Yield curve changes & the 5-step return decomposition · Static curve strategies (view = market's forward view) · Divergent view on level · Divergent view on slope · Divergent view on curvature: butterfly trades · Key rate durations · Across currencies: unhedged, hedged, carry · Evaluating a yield curve strategy (worked)

Ch 6 · Credit risk, spreads and the credit cycle · Fixed-rate spread measures · Spread duration, DTS and excess spread · Bottom-up credit: models & relative value · Liquidity risk & tail risk (VaR) · Credit default swaps: pricing · CDS long/short and curve trades · Static & dynamic credit curve strategies

Ch 7 · Why managers trade · Inputs to trading strategy & the trader's dilemma · Reference price benchmarks · Execution methods & algorithms · Implementation shortfall (worked) · Evaluating execution: arrival, market-adjusted, added value · Trade governance & best execution

Ch 8 · How to read this case · Managing liquidity risk: four tools · QU endowment: facts you need · Proposed allocation: justify it & name the trade-offs · Why liquidity gets worse in a crisis · Ethics in manager selection (who broke what) · Equity exposure: ETF vs futures vs total return swap · Rebalancing: cash market or derivatives? · ESG policy & the child-labour case

Ch F · Equity (P1–P3) · Liability-driven & yield curve (P4–P5) · Credit (P6) · Trading (P7)

4★ list (19)

Ch 1 · Factor-based vs market-cap indexing · Ways to get index exposure · Tracking error: causes, control, attribution

Ch 2 · Top-down strategies · Building a fundamental strategy & its pitfalls

Ch 3 · Approaches, objectives and constraints · Risk budgeting: relative (active) risk · Choosing the level of risk · Heuristic vs formal risk constraints · The well-constructed portfolio

Ch 4 · LDI risks · Passive exposure alternatives & benchmark choice

Ch 5 · Volatility views: embedded & stand-alone options

Ch 6 · Floating-rate note spreads · Top-down, factor and ESG credit strategies · Global credit, emerging markets · Structured credit & fixed-income analytics

Ch 7 · Trading by market

Ch 8 · The illiquidity premium

Source: Kaplan Schweser 2025 CFA Level III Portfolio Management SchweserNotes (readings P1–P8 and formula sheet). Weight estimates: CFA Institute's 30–35% pathway share, split by curriculum depth. The core readings (asset allocation, portfolio construction, performance, derivatives & risk, ethics) are not in this file. CFA® is a trademark of CFA Institute.

00 Exam Trend & How to Use This Summary

≈— marks · 3 subtopics · chapter priority ★★★★★

Level III exam structure (Feb 2027) ★★★★★

Item	Detail
Question sets	11 item sets + 11 essay (constructed response) sets, 12 points each; 20 scored, 2 trial
Item set	Short vignette + 4 multiple-choice questions
Essay set	Vignette + questions you answer in writing: justify, calculate, 'determine and justify'
Core share	≈ 65–70% (asset allocation, portfolio construction, performance, derivatives & risk, ethics)
PM pathway share	≈ **30–35%** ≈ 7 scored sets

TRAP Topic weights here are from CFA Institute's 2025 pathway structure and prep providers. Check the official Level III page for any 2027 change.

Where the pathway marks come from ★★★★★

Reading	Share of pathway (est.)	Exam style	Priority
P6 Credit strategies	≈ 19%	Calculations (spread measures, CDS), strategy choice	★★★★★
P8 Institutional case study	≈ 15%	Essay: liquidity, asset allocation changes, ESG, risk	★★★★★
P4 LDI & index-based fixed income	≈ 14%	Calculations: immunisation, BPV hedge, contingent immunisation	★★★★★
P5 Yield curve strategies	≈ 13%	'Which trade?' + return estimates + currency	★★★★★
P2 Active equity: strategies	≈ 12%	Identify strategy / style / IC	★★★★
P7 Trade strategy & execution	≈ 12%	Benchmarks, implementation shortfall, algo choice	★★★★
P3 Active equity: portfolio construction	≈ 11%	Active share vs active risk, risk budget, long/short	★★★★
P1 Index-based equity	≈ 4%	Short explain questions	★★★

Fixed income (P4 + P5 + P6) ≈ **46%** of the pathway — prep providers call it the heaviest technical content

Essays favour 'determine the most appropriate strategy and justify'

Numbers to master: BPV hedge ratio, immunisation, rolldown return, spread duration, CDS price, implementation shortfall

How to read this summary ★★★☆☆

- Each subtopic starts with **In plain words** — read that first
- Formula boxes show the formula, then a worked example from the notes
- Red TRAP boxes are the mistakes that cost marks
- Page 1 lists every 5★ subtopic — the minimum to study

01 P1 Index-Based Equity Strategies

≈4% of PM marks · 4 subtopics · chapter priority ★★★☆☆

Factor-based vs market-cap indexing ★★★★☆

In plain words: a normal index fund buys every stock in proportion to its size (market cap). A **factor-based** fund instead tilts toward stocks with a chosen trait — cheap, small, high momentum — because the investor believes that trait will earn more.

- CAPM = one factor (market beta); multifactor models = several
- Fama–French (2015) five factors: market risk premium · size · book-to-market (value) · operating profitability · investment intensity (asset growth)
- Common factors: growth, value, size, yield, momentum, quality, volatility

Type	Examples
Return-oriented	Dividend yield, momentum, **fundamentally weighted** (dividends, sales, income)
Risk-oriented	Volatility weighting (weights = 1/volatility), minimum-variance (Markowitz with constraints)
Diversification-oriented	Equal weighting, maximum diversification (max ratio of weighted-average stock volatility to portfolio volatility)

- Factor funds are rules-based, transparent → 'passive', but tilting is a form of active management
- Cheaper than active management, dearer than cap-weighted index funds
- Transparency lets others copy → may erode the extra return
- Multiple benchmarks (factor + cap-weighted) can raise tracking error
- Risk-oriented: simple, reduce risk, but rely on past data

> **TRAP** A small-cap, high-P/E (growth) strategy is **RETURN-ORIENTED**, not risk-oriented.

Ways to get index exposure ★★★★☆

Approach	Pros	Cons
Pooled: open-end mutual fund	Low transaction cost, convenient	Must sell stocks to meet redemptions → taxable gains
Pooled: ETF	Trades intraday; in-kind redemptions → no forced sales, fewer taxable gains	Commissions, bid-ask spread, some ETFs illiquid
Derivatives (futures, options, swaps)	Quick, cheap exposure change; liquid; easy leverage	Expire → must roll; position limits; may not fit special needs; OTC counterparty risk; basis risk → tracking error
Separately managed portfolio	Full control	Needs index data, trading & accounting systems, broker relationships (program trading), compliance systems

Overlay	Purpose
Completion overlay	Bring portfolio back to index risk (e.g. match beta)
Rebalancing overlay	Cheaply match index reconstitution
Currency overlay	Adjust FX risk of foreign holdings

Full replication, stratified sampling, optimisation ★★★★★

Method	How	Best for / drawbacks
Full replication	Hold every index stock at index weight	Closest match before costs; costly for many/illiquid stocks; best for **few, liquid, similar** stocks
Stratified sampling	Split index into mutually exclusive, exhaustive strata (size, sector, country, yield…); sample within each to match factor exposures	More criteria → lower tracking error
Optimisation	Mean-variance: minimise tracking error subject to constraints	Accounts for covariances; lower TE than sampling; but based on history, costly to maintain, may not be mean-variance efficient
Blended	Replicate large liquid names, sample/optimise the thin tail	Broad indexes like Wilshire 5000

> **TRAP** As you add stocks, tracking error **FIRST FALLS, THEN RISES**: the later, less liquid stocks add trading costs.

Tracking error: causes, control, attribution ★★★★☆

In plain words: tracking error is how much the fund's return wobbles around the index return (SD of the return differences). For an index fund, low tracking error — not high excess return — shows skill.

- Causes: management fees · commissions · sampling · intraday trading (index uses closing prices) · **cash drag**
- Control: trade-off between replication cost and sampling error; use **index futures** to equitise cash
- Attribution analysis finds the sources of tracking error
- **Securities lending** income offsets fees → lowers tracking error
- Index managers still owe a fiduciary duty to vote proxies; many use proxy-voting services

02 P2 Active Equity Investing: Strategies

≈12% of PM marks · 9 subtopics · chapter priority ★★★★☆

Fundamental vs quantitative approaches ★★★★★

In plain words: both try to beat a benchmark. A **fundamental** manager decides by **judgment** after studying companies. A **quantitative** manager decides by **rules** built from data. The difference is how the buy decision is made, not the tools used.

	Fundamental	Quantitative
Style	Subjective, analyst discretion	Objective, systematic rules from data
Focus	Company: business model, management, valuation	Factors across many stocks
Portfolio	Fewer, larger, high-conviction positions	Many small positions; optimiser maximises alpha / IR
Main risk	Company level: intrinsic value wrong, or mispricing never corrects	Portfolio level: factors stop working
Rebalancing	Continuous, by opinion	Fixed intervals (monthly, quarterly), by rules

TRAP Screening for low P/B can be used by BOTH types. What matters is how the final decision is made.

Bottom-up strategies: value & growth sub-styles ★★★★★

Value sub-style	Looks for
Relative value	Low P/E or P/B vs peers, without good reason
Contrarian	Against sentiment — depressed cyclicals with low/negative earnings
High-quality value	Value + financial strength + good management + profitability ('Warren Buffett')
Income	High dividend yield, dividend growth
Deep value	Very low value vs assets (low P/B), often distressed
Restructuring & distressed debt	Before/during bankruptcy — value released by restructuring or liquidation
Special situations	Mispricing from divestitures, spin-offs, mergers

Growth sub-style	Looks for
Consistent growth	Long-term steady growth
Earnings momentum	Short-term earnings acceleration
GARP	Growth at reasonable price; **PEG = P/E ÷ growth %**

```
Matching example: lowest P/B 0.75 → deep value · lowest PEG (15/10 = 1.5) → GARP
Lowest P/E vs sector (20 vs 30 = 0.67) → relative value · highest yield (1.08/12 = 9%) → income
```

Fundamental bottom-up: strong business model, brand, competitive advantage, management, governance

Undervalued vs liquidation value → **distressed debt**, not deep value

Top-down strategies ★★★★☆

- Macro, policy and market-level views; implement with broad ETFs and derivatives
- Dimensions: **country/geography · industry sector · volatility** (VIX futures, variance swaps, straddles) · **thematic** (cloud, blockchain, clean energy; or short-term like a political vote)
- Blend: top-down sets country/sector weights, bottom-up picks stocks; or bottom-up portfolio + top-down derivatives overlay to remove unintended macro bets
- 'Smart beta' = rotating into factors expected to outperform — a top-down idea

TRAP A manager hunting GARP among individual S&P 500 stocks is BOTTOM-UP, even if the universe is an index.

Factor-based active strategies & the hedged portfolio ★★★★★

In plain words: a factor is a stock trait linked to returns (small size, cheapness). To measure a factor, rank stocks by it, buy the best slice and short the worst slice, and track that long/short portfolio.

```
Hedged portfolio (Fama-French): rank universe by factor → split into quantiles (deciles/quintiles)
→ long best quantile, short worst → track return = factor return
Value factor: long HIGH book-to-market (cheap), short LOW book-to-market (expensive)
```

Drawback of hedged portfolio approach	
1	Ignores middle quantiles
2	Assumes a linear factor–return link
3	Correlated factors give less diversification than it seems
4	Needs short selling
5	Not 'pure' — has exposures to other factors

Rewarded factors = persistent premium; avoid unrewarded ones; factor must make intuitive sense (else data-mined)

Factor-mimicking portfolio: dollar-neutral, exposure 1 to one factor, 0 to others — hard to build (liquidity, shorting)

Long-only investors tilt toward factors; small tilts = enhanced indexing

Factor timing / **style rotation**: regress factor returns on drivers (e.g. rate surprises)

```
Rate-surprise regression: current month beta coefficient -0.5 (high beta underperforms after upward surprise)
Next month: size -1.0 (small caps underperform), value negative, beta ≈ +0.7 → timing of the beta effect needs study
```

TRAP Using quartiles instead of deciles is NOT an error. Buying low book-to-market and shorting high IS the error.

Activist strategies ★★★★★

```
Process: screen → buy stake (usually < 10% of votes) → public proposal (open letter) → threaten proxy contest → launch proxy contest / keep negotiating
```

Activist tactics	Management defences
Seek board seats; open letters; rally shareholders	Multi-class shares (founder super-votes)
Proposals at AGM	**Poison pill** (holders buy cheap shares, diluting activist)
Higher dividends, buybacks; cut excessive pay	**Staggered board** (partial re-election each year)
Lawsuits for breach of fiduciary duty; media campaigns; break up conglomerates	

Roots: 1970s–80s corporate raiders; activist AUM more than doubled 2007–2015, campaigns ×4

Targets: slow growth, negative momentum, weak governance; may be ESG-driven

Impact: better growth, profitability, governance — but **higher debt**; Sharpe slightly above market; prices rise into and on announcement day

Activists do **not** take majority stakes

```
Case: Formby (cash-rich, no dividend, slowing growth) → activist (push payouts/buybacks)
```

Statistical arbitrage & event-driven strategies ★★★★★

Strategy	How	Risk
Pairs trading (stat arb)	Two historically correlated stocks; buy the laggard, short the leader; spread = ln(price ratio); sell spread > +2 SD of moving average, buy < −2 SD	Relationship broke for good → no mean reversion
Market microstructure arbitrage	Millisecond order-book imbalances; needs HFT	Technology/latency
Event-driven	Mispricing around M&A, earnings, restructuring, buybacks, special dividends, spin-offs (no activism)	Event outcome
Risk (merger) arbitrage	Cash deal: buy target below deal price. Stock deal: long target + short acquirer at exchange ratio	Deal fails → large loss; skill = estimating failure probability, timing, annualised premium

```
Case: Parmeon (1-for-1 share deal, regulatory doubt) → event-driven: buy target, short Parmeon 1:1
Baron (profit warnings, new management, debt doubts, correlation broke) → distressed debt, NOT pairs trade
```

Building a fundamental strategy & its pitfalls ★★★★☆

```
1. Define universe per mandate + investment thesis
2. Prescreen (e.g. drop high P/E)
3. Analyse industry, competitive position, financials
4. Forecast cash flows / earnings
5. Convert forecasts to valuations
6. Construct portfolio with desired risk (add top-down views)
7. Rebalance; sell disciplines: target prices, stop-losses
```

Biases: confirmation, illusion of control, availability, loss aversion, overconfidence, regret aversion

Value trap: cheap for a reason — fundamentals deteriorating; need a catalyst

Growth trap: growth already in the price (priced for 15%, delivers 12% → P/E collapses)

Building a quantitative strategy; information coefficient ★★★★★

```
1. Define market opportunity  2. Acquire & process data (most time-consuming)  3. Backtest  4. Evaluate (out-of-sample, VaR, drawdown)  5. Construct portfolio (risk model + trading costs)
Standardised factor score = (value - mean) / SD   e.g. E/P 7%, mean 3%, SD 5% → 0.8
Pearson IC = correlation(factor score, SUBSEQUENT return); 5–6% monthly = very strong
Spearman rank IC = correlation of ranks → robust to outliers
```

```
9-stock example: Pearson IC -0.99% (distorted by outlier stock A); Spearman +40% → E/P predictive
Long top 3 (G, H, I) = (3.02 + 3.05 + 5.20)/3 = 3.76% ; short bottom 3 (A, B, C) = 2.99% → L/S = 0.77%
Small sample → retest on more data
```

Data types: company mapping · fundamentals · survey (estimates, sentiment, flows) · unconventional (satellite, news sentiment, supply chains)

Pitfalls: **survivorship** bias · **look-ahead** bias (Dec accounts used in Jan) · data mining / overfitting · turnover · stock not available to borrow · transaction costs

Quant overcrowding: everyone exits together (August 2007); warning sign = high correlation of short interest and momentum

Correlated factors add little in a multi-factor model

TRAP IC must use **SUBSEQUENT** returns, not contemporaneous. Worried about outliers → Spearman rank IC.

Style classification: holdings-based vs returns-based ★★★★★

- Styles should be highly correlated within, low correlated between
- **Holdings-based** (Morningstar style box): size (large ≈ top 70% of cap, mid next 20%, small rest) × value/blend/growth; each stock gets value and growth scores 0–100; net = growth − value: strongly negative → value, positive → growth, near zero → core (stock) / blend (fund)
- **Returns-based** (Sharpe): regress fund returns on style indexes, coefficients sum to 1 = style allocation (e.g. large value 0.85, small value 0.15, growth 0)
- **Self-identification**: check against both; long/short, market-neutral, short-bias funds rely on prospectus

	Holdings-based	Returns-based
Pros	Shows **current** exposure; precise; spots drift	Little data needed; captures **average** exposure over time; picks up derivatives overlays
Cons	Needs frequent holdings data; misses undisclosed **derivatives overlays**; data-intensive	Needs long history; results depend on chosen indexes; shows past, not current, style

TRAP Derivatives overlays with limited disclosure hurt **HOLDINGS-BASED** analysis only.

03 P3 Active Equity Investing: Portfolio Construction

≈11% of PM marks · 11 subtopics · chapter priority ★★★★☆

Sources of active return & the building blocks ★★★★★

In plain words: active return = portfolio return − benchmark return. It is positive if you overweight the winners. It comes from three places: different exposure to rewarded factors, true skill (alpha), and plain luck from concentrated bets.

```
Active return = Σ(βpi - ββi) × Factor return i   +   α (skill)   +   ε (idiosyncratic = luck)
```

Building block	Meaning	Link to return source
1. Factor weightings	Over/underweight rewarded factors (market, size, value, momentum, liquidity) vs benchmark; static exposure no longer counts as alpha	Long-term factor exposure
2. Alpha skills	Factor timing (rewarded or unrewarded: sectors, regions, commodities), security selection	Mispricing → alpha
3. Position sizing	Balance conviction vs idiosyncratic risk; more, smaller positions → lower risk	Mostly idiosyncratic risk
4. Breadth of expertise	Ties them together — more independent decisions	Consistency

Fundamental law of active management ★★★★★

```
E(R_A) = IC × √BR × σ_A × TC
IC = information coefficient (skill), BR = breadth (independent decisions per year),
σ_A = active risk, TC = transfer coefficient (how well insights reach the portfolio)
Example: 10 decisions/month → BR = 120; IC 0.1, σ_A 5%, TC 0.5
E(R_A) = 0.1 × √120 × 5% × 0.5 = 2.74%
```

One factor, one metric → decisions not independent → low breadth

Shorting / fewer constraints raise TC → higher IR

TRAP Convert breadth to **ANNUAL** : 10 a month = 120 a year.

Approaches, objectives and constraints ★★★★☆

Approach	Typical use of building blocks
Systematic (rules)	Diversified factor exposures, optimisation, many positions
Discretionary (judgment)	More factor timing, concentrated, less formal optimisation
Bottom-up / top-down	Security information vs macro information; factor timing mostly **discretionary top-down**

	Absolute	Relative to benchmark
Objective	Maximise Sharpe ratio	Maximise information ratio
Constraint	Max volatility	Max active risk / tracking error

Other objectives: minimise volatility/downside/drawdown · maximise rewarded-factor exposure · maximise custom traits (deep value) · heuristic ranking (e.g. by low P/B)

Active Share vs active risk ★★★★★

```
Active Share = ½ Σ |wpi - wβi|        0 = index; 1 = no overlap
1 - Active Share = overlap with benchmark
Active risk (tracking error) = SD of (Rp - Rβ)
Active risk² = active factor risk (β differences) + active idiosyncratic risk (incl. alpha)
```

	Active Share	Active risk
Measures	Holdings difference from benchmark	Volatility of active returns
Correlation matters?	No	Yes
Manager control	**Full** (sets weights)	Partial (depends on estimated variances/correlations)
Swap pharma A for similar pharma B	↑	Barely ↑
Swap pharma for consumer stock	↑	↑

Fewer, more concentrated stocks → higher Active Share

Active Share of 0.2 at the same fee as 0.5 = expensive (closet indexing)

High net factor exposure → high active risk; no factor exposure → active risk all from Active Share (falls as number of stocks rises)

Manager type	Active Share	Active risk
Closet indexer	Low	Low
Diversified multi-factor	Moderate	Moderate
Sector rotator	Moderate	High (large sector deviations)
Concentrated stock picker	High	High (large single-stock deviations, sector-neutral possible)

TRAP Factor bets always raise Active Share, but high Active Share need not mean factor bets (two stocks with identical factor exposures).

Risk budgeting: absolute risk ★★★★★

In plain words: a risk budget hands out the total risk you are willing to take across positions or factors, like a spending budget.

```
4 steps: choose measure (absolute for total-return mandates, relative for index-beating) → understand risk sources → set budget → allocate
Contribution of asset i to variance: CVi = Σj wi wj Cov(i,j) = wi × Cov(i, portfolio)
Asset A (w 0.20, var 0.05): 0.2×0.2×0.05 + 0.2×0.3×0.0067 + 0.2×0.5×0.0013 = 0.002532
Portfolio variance = 0.002532 + 0.001998 + 0.002880 = 0.00741 → A's share = 34.2%
Factor contribution: Σj βi βj Cov(Fi,Fj); market: 0.892²×0.00178 + 0.892×(-0.283)×0.00042 = 0.00131
Total variance 0.0374² = 0.0013988 → market explains 93.7%; size -0.00007 → unexplained = 0.00016 = 11.4%
```

Contribution of a sector/country = its weight × covariance with the **portfolio**

Shares can exceed 100% when a factor contributes negatively

Risk budgeting: relative (active) risk ★★★★☆

```
Contribution of asset i to active variance: CAVᵢ = (wₚᵢ - wᵦᵢ) × Σⱼ (wₚⱼ - wᵦⱼ) × Cov(active returns i, j)
```

Driven by active weight and active risk, not absolute SD

50/50 benchmark of two indexes + cash in portfolio: cash's active risk is high (low correlation) → cash = 100% of active variance; the two indexes' active returns correlate −1

Choosing the level of risk ★★★★☆

- Implementation constraints (short/leverage limits, liquidity costs) → IR degrades as active risk rises
- Limited diversification at high risk → Sharpe falls (concave frontier)
- Leverage: $R_g \approx R_a - \sigma^2/2$ → too much leverage lowers compounded return (−20% then +20% = 0.96)

Heuristic vs formal risk constraints ★★★★☆

Type	Examples	Needs return forecasts?
Heuristic	Position, sector, region limits; leverage, liquidity, turnover limits	No
Formal	Volatility, active risk, skewness, drawdown, VaR, **CVaR** (expected loss beyond VaR), **IVaR** (VaR change from adding a position), **MVaR** (VaR change from a tiny position change)	**Yes** → estimation error

Leverage magnifies estimation errors; tail events force sales

Absolute-return managers watch drawdown; market-neutral managers target absolute volatility

Concentrated portfolios → larger estimation error

AUM, liquidity, turnover and market impact ★★★★★

```
Small cap $5bn, 1% trades daily → ADV $50m; limit 10% of ADV → max $5m
In a $1bn fund → max weight 0.5%
Scale example: $20tn index, min index weight 0.02% → smallest stock $4bn; ADV 1.5% → $60m
Max 5% of ADV → $3m; max weight = min(10 × 0.02%, 0.02% + 1%) = 0.2%
Strategy impaired when AUM > $3m / 0.002 = $1.5bn
```

Impact cost ↑ with: higher AUM vs market cap · smaller caps · higher turnover · shorter horizon · information-rich trades

Slippage = execution price − midquote when order entered: usually > explicit costs · higher for small caps · not necessarily higher in emerging markets · much higher in volatile markets

Small-cap managers: close fund, diversify, cut turnover or change trading; beware track records that can't scale

The well-constructed portfolio ★★★★☆

Clear philosophy, consistent process · promised risk characteristics · efficient exposures · reasonable costs

Same exposures with fewer positions → more risk-efficient

Same factors → prefer lower absolute and active risk

Same risks, costs, skill → prefer **higher Active Share**

Combine managers to fill missing factor exposures

Long-only, long/short, long extension, market-neutral ★★★★★

```
Gross exposure = longs + |shorts| ; net = longs - shorts (as % of capital)
$100m capital, $80m long, $30m short → gross 110%, net 50%, cash $50m ($20m + $30m short proceeds)
130/30 long extension: net 100%, gross 160%
Market-neutral: $200m long β 0.9 (=180) vs $150m short β 1.2 (=180) → net beta 0
```

Long-only merits	Long/short benefits	Long/short drawbacks
Earns long-term risk premiums	Express negative views fully (long-only can at most not hold) → higher TC/IR	Unlimited loss on shorts; gives up market premium
Higher capacity (liquidity vs borrow availability)	Use short proceeds to gear into best longs	Leverage magnifies losses
Limited liability — loss capped	Remove market risk → diversifier	Borrow costs can spike
No short-sale bans, simpler (no prime broker, collateral, recall risk)	Better control of factor exposures	Collateral calls, **short squeeze**, lender recalls
Lower fees and costs; ideology		

Market-neutral built via pairs trading (quant = stat arb); low volatility, low correlation, mostly for diversification; hard to hold beta at zero

04 P4 Liability-Driven & Index-Based Fixed Income

≈14% of PM marks · 8 subtopics · chapter priority ★★★★★

Immunisation & duration basics ★★★★★

In plain words: immunisation means building a bond portfolio that will be worth enough to pay a known future bill, whatever happens to interest rates. The easiest way is one zero-coupon bond maturing on the bill date. Otherwise you match **duration**, so price risk and reinvestment risk cancel.

Macaulay duration = PV-weighted average time to cash flows; zero-coupon = its maturity

If rates rise: price loss offset by higher reinvestment income (and vice versa) — for one small, immediate, parallel shift

Modified duration = Macaulay ÷ (1 + periodic yield) → better for immediate price change; Macaulay → better for immunisation timing

Use **portfolio statistics from aggregate cash flows** (portfolio IRR, duration, dispersion, convexity), not market-value weighted averages; same only if curve is flat; with upward slope the cash-flow IRR and duration are higher

Goal = earn the initial **portfolio IRR**, not average YTM

```
%ΔValue ≈ -MD × ΔY + ½ × C × (ΔY)²
BPV (PVBP, 'value of an 01') = MD × Value × 0.0001
```

Single-liability immunisation rules & structural risk ★★★★★

```
1. PV assets ≥ PV liability
2. Macaulay duration of assets = liability due date
3. Minimise convexity (least dispersion of asset cash flows around the liability)
4. Rebalance regularly (trade-off with transaction costs)
```

Curve move (barbell assets around bullet liability)	Effect
Parallel shift up or down	Assets beat liability (asset convexity > liability) → OK
Steepening twist	PV assets falls below PV liability → risk (may still work if IRR rises enough)
Flattening twist	PV assets rises but IRR falls → risk
Positive butterfly (wings up, middle down)	PV assets ↓, PV liability ↑ → significant risk
Negative butterfly	PV assets ↑, PV liability ↓ → favourable but not guaranteed

Immunisation = **zero replication**: success if portfolio IRR moves like the replicating zero's yield

Parallel shift is **sufficient but not necessary** for success

Not buy-and-hold: coupon bond duration falls slower than the liability's → rebalance

TRAP Minimise convexity for a single liability — positive convexity is good for price, but more dispersion means more structural risk.

Multiple liabilities: cash-flow vs duration matching ★★★★★

Method	How	Notes
Cash-flow matching (dedication)	Bonds whose coupons/principal match each payment	Can legally **defease** (remove from balance sheet) with govt bonds; costly
Duration matching	Match **BPV** of assets and liabilities	Cheaper, flexible; structural risk
Contingent immunisation	Manage actively while surplus > 0; immunise if surplus hits zero	Needs a significant surplus

```
Multiple-liability rules: 1. PV A ≥ PV L  2. BPV A = BPV L  3. Asset dispersion & convexity exceed liabilities (but not by much)  4. Rebalance
```

```
UK company: Portfolio B — BPV closely matches, convexity slightly above liabilities → best
Expect very large parallel shifts → Portfolio C (much higher convexity) despite weaker BPV match
```

TRAP Single liability: minimise convexity. Multiple liabilities: asset convexity must be **GREATER** than liability convexity, but minimised subject to that.

Closing a duration gap with futures ★★★★★

```
Futures BPV ≈ BPV_CTD ÷ CF_CTD
Number of contracts = (BPV_L - BPV_A) ÷ BPV_futures  → positive = BUY
UK: gap = 218,517 - 203,456 = 15,061
5-yr: BPV = 48.1650/0.9237 = 52.14 → buy 289 ; 10-yr: 86.7001/0.9169 = 94.56 → buy 159
Liability duration 8.99 → 10-yr contract is the closer match → less structural risk
```

```
Pension: PBO $2.57bn, duration 9.35 → BPV_L = 2.57bn × 9.35 × 0.0001 = 2,402,950
Assets $3.07bn: 60% equity (duration 0) + 40% bonds duration 2.85 → BPV_A = 1.228bn × 2.85 × 0.0001 = 349,980
Gap = 2,052,970 ; CTD BPV 128.98, CF 0.9436 → contract BPV 136.69
100% hedge: buy ≈ 15,019 ; expect rates ↓ → overhedge (110%) ≈ 16,521 ; expect rates ↑ → underhedge (90%) ≈ 13,517
```

CTD = cheapest-to-deliver; seller delivers 100,000 par, buyer pays price × conversion factor; Ultra 10-year narrows deliverables

Each 1/32 move = $31.25 per contract → daily margin cash flows → 100% hedges are rare in practice

Equity duration assumed zero

ABO (legal, if closed now) < PBO (ongoing plan — usually more realistic)

TRAP Assets' BPV below liabilities' → **BUY** futures / receive fixed. Expect rates to rise → hedge **LESS** ; expect rates to fall → hedge **MORE** .

Swaps, swaptions and collars for the duration gap ★★★★★

```
Swap BPV per 100 notional = (D_fixed - D_floating) × 100 × 0.0001
Notional = BPV gap ÷ (swap BPV / 100)
Pension: (9.18 - 0.25) × 0.01 = 0.0893 → NP = 2,052,970 / 0.000893 ≈ $2.30bn receive-fixed
Range 30–70%: expect rates ↑ → 30% = $0.69bn ; then expect ↓ → 70% = $1.61bn (add $0.92bn)
Quiz: gap $300,000, D 8.25 and 0.5 → BPV 0.0775 → NP = $387m
```

```
US plan: gap 450,000, swap BPV 0.2571 per 100 → 50% hedge = 450,000 × 0.5 / 0.002571 = $87.5m receive-fixed
Want 35% if rates rise → buy payer swaption on 15% = $26.25m; premium 55 bp = $144,397
Exercise if new swap rates > 2.7% strike
```

Need more asset duration — choice	Optimal when new swap fixed rate (SFR) is...
Receive-fixed swap at 2.5%	≤ 2.5% (rates fall)
Collar: buy 2.3% receiver + sell 3.3% payer (zero cost)	Between 2.5% and 3.3%
Buy 2.3% receiver swaption	Well above 3.3% (cost capped at premium)

- Receive-fixed = + duration; pay-fixed = − duration
- Payer/receiver always describe the **buyer's** fixed leg
- Swaption premium is sunk — ignore it in the exercise decision
- Swaps now often margined → same cash-flow issues as futures

LDI risks ★★★★☆

- Duration-only hedges ignore convexity
- Duration assumes parallel shifts
- Curve twists → structural risk
- Model risk in estimating DB liabilities
- Measurement error from weighted averages
- CTD can change → futures BPV changes
- **Spread risk**: asset (govt/swap) rate vs liability (corporate) discount rate move differently
- Counterparty risk (OTC)
- Collateral/margin cash-flow risk
- Asset liquidity risk

Bond indexing: why it's hard & how to match ★★★★★

- Bond market huge, issues varied → full replication impractical
- Issuers have many issues of different liquidity
- OTC dealer market; post-2008 capital rules → less inventory, wider spreads
- Most bonds don't trade → **matrix / evaluated pricing**
- Index composition changes quickly

Match	Controls
Modified (effective, if options) duration	Parallel shifts
Key rate durations / PV distribution of cash flows	**Non-parallel** shifts
Sector & quality weights; spread duration	Spread changes
Sector/coupon/maturity cells; callable/MBS weights	Convexity, options
Issuer exposure	Event risk

- Enhanced indexing: match risk factors, not holdings; **stratified sampling / cell matching**; can include ESG exclusions/inclusions
- Add value: cut costs · tilt to cheap sectors/curve points (avoid spread widening) · overweight callables when rate volatility expected low

Passive exposure alternatives & benchmark choice ★★★★☆

Vehicle	Points
Separately managed (full/enhanced index)	Control; high cost
Index mutual fund	Small investors; daily NAV; fees, loads; never matures
ETF	Trades all day; authorised participants create/redeem in kind → arbitrage (weaker for illiquid bonds)
Total return swap	Receive index total return, pay MRR + spread = buying index with borrowed money; leveraged if not collateralised; counterparty risk; negative returns paid by receiver

- Start from objectives/constraints → strategic allocation → index
- Static portfolio duration falls as bonds age
- Issuance shifts change index duration
- Custom 'smart beta' benchmark, e.g. **credit barbell**: long Treasuries for duration + short corporates for spread

05 P5 Yield Curve Strategies

≈13% of PM marks · 9 subtopics · chapter priority ★★★★★

Yield curve changes & the 5-step return decomposition ★★★★★

In plain words: the yield curve plots government bond yields against maturity. It moves in three ways — up/down (**level**), steeper/flatter (**slope**), more/less bent (**curvature**). Level shifts explain most bond price changes.

```
Slope = long yield - short yield (rising = steepening)
Butterfly spread = -short + 2 × medium - long  (rising =
more curvature = NEGATIVE butterfly)
1. Coupon income = annual coupon / price
2. Rolldown = (price with unchanged curve - today's
price) / today's price
3. Yield change = -MD × ΔY + ½ C × ΔY²
4. Spread change = -MD × ΔS + ½ C × ΔS²
5. Currency gain/loss (compound: (1 + R_local)(1 + R_FX)
- 1)
```

Curves built from liquid on-the-run issues (off-the-run would give a higher curve — liquidity premium); interpolation and bond choice matter

Higher convexity → better in volatile rates → trades at a premium (lower yield)

Barbell (dispersed cash flows) has more convexity than a bullet of the same duration

```
Bullet 10-yr (MD 9.52, C 104.8) vs barbell 46.68% 2-yr
(1.86, 5.2) + 53.32% 20-yr (16.23, 292.8), same duration
9.52
+50 bp: bullet -4.63% ; barbell 0.4668×(-0.92%) +
0.5332×(-7.75%) = -4.56% → barbell wins on convexity
```

TRAP A **NEGATIVE** butterfly twist = curvature **UP** = butterfly spread **UP** (the curve 'frowns' more).

Static curve strategies (view = market's forward view) ★★★★★

Strategy	How	Main risk
Buy and hold	Extend duration for higher coupons; low turnover	Rates rise (parallel shift up)
Rolling down the curve	Buy at the end of a **steep** segment; yield falls as it ages → price gain; sell and roll out	Curve shifts/flattens
Repo carry trade	Borrow short (repo) to buy long bonds	Rates rise; funding
Long futures	Implicit leverage via margin	Rates rise
Receive-fixed swap	Earn carry = swap fixed rate − MRR + mark-to-market	Rates rise

```
Bond vs swap, £50m, 6 months, rates -25 bp
Trade A: 10-yr 3% gilt at 104.3998 (yield 2.5%)
Coupon = 1.5% × 50m = £750,000 ; new price (19 periods,
1.125%) 106.3828 → +£991,500
of which rolldown (price at old yield 104.2048) = -
£97,500 ; rate change = +£1,089,000
Trade B: receive 3%, pay MRR 0.5% → carry = (3% - 0.5%)/2
× 50m = £625,000
Fixed leg at 2.75%: 1.020776 × 50m = 51,038,817 → MTM = +
£1,038,817 (floating leg resets to par)
```

Futures contracts = target BPV ÷ (BPV_CTD / CF_CTD)

LIBOR replaced by market reference rates (MRR)

TRAP Stable-curve manager adds duration: **RECEIVE-FIXED** swap, **REPO** (borrow), not reverse repo.

Divergent view on level ★★★★★

In plain words: expect rates to fall (**bullish**) → lengthen duration. Expect rates to rise (**bearish**) → shorten duration.

Tool	Increase duration	Decrease duration
Cash bonds	Buy long bonds	Buy short bonds / sell long
Futures	Buy	Sell
Swaps	Receive fixed	Pay fixed

```
Benchmark equal-weight 2/10/20-yr: D 9.203, C 134.3 ;
bearish portfolio 50/25/25: D 7.368, C 102
+40 bp: benchmark -3.57%, portfolio -2.87% → excess
+0.70%
Same via swap: BPV target 73,680 - 92,030 = -18,350 ;
pay-fixed swap MD -8.354
NP = -18,350 / (-8.354 × 0.0001) = $21.98m
```

Divergent view on slope ★★★★★

In plain words: buy bonds whose yields you think will fall relative to others; short bonds whose yields will rise. Set overall duration by your level view: positive if bullish, negative if bearish, zero if neutral.

View	What happens	Position
Bull steepener	Short rates fall more (central bank cuts in recession)	Long short end; duration positive
Bear steepener	Long rates rise more (CB too loose → inflation fears)	Short long end; duration negative
Bear flattener	Short rates rise more (CB hikes)	Short short end; duration negative
Bull flattener	Long rates fall more (flight to quality)	Long long end; duration positive

```
Steepener, duration-neutral: long £100m 2-yr → BPV 100m ×
1.86 × 0.0001 = £18,600
Short 10-yr MV = 18,600 / (9.52 × 0.0001) = £19,537,815
2-yr -20 bp: +0.373% = +£373,040 ; 10-yr +20 bp: -1.883%
→ short gains £367,903 → total ≈ +£740,900
Bull steepener next → tilt to positive duration (add 2-yr
or short less 10-yr)
```

TRAP Negative duration + long short-term + short long-term = **BEAR STEEPENER** .

Divergent view on curvature: butterfly trades ★★★★★

View	Means	Trade
Curvature ↑ (negative butterfly)	Middle yields rise vs wings	**Short body (bullet), long wings (barbell)**
Curvature ↓ (positive butterfly)	Middle yields fall vs wings	**Long body, short wings**

```
Positive butterfly, duration-neutral, equal wings: long
£100m 10-yr → BPV £95,200
MV_w × (1.86 + 16.23) × 0.0001 = 95,200 → MV_w =
£52,625,760 each wing short
Wings +10 bp, body -25 bp: body +2.413% (£2.41m) ; short
2-yr gains ≈ £0.10m ; short 20-yr gains ≈ £0.85m → ≈ +
£3.36m
```

Butterflies are duration-neutral but **not** convexity-neutral: long body/short wings = short convexity

Volatility views: embedded & stand-alone options ★★★★☆

Instrument	Effect on duration	Benefits when
Callable bond (bond + short call)	Lower; negative convexity at low yields	Rates stable/rising, low volatility
Putable bond (bond + long put)	Lower; price floor at high yields	Rates rise / volatility high
Long call on bond or bond futures	↑	Rates fall
Long put on bond or futures	↓	Rates rise
Long receiver swaption	↑	Rates fall
Long payer swaption	↓	Rates rise

$$\text{Effective duration} = (PV_- - PV_+) / (2 \times \Delta\text{Curve} \times PV_0)$$
$$\text{Effective convexity} = (PV_- + PV_+ - 2PV_0) / (\Delta\text{Curve}^2 \times PV_0)$$

- Use effective (not modified) measures for bonds with options
- Embedded options always **lower** duration
- Expect rates up → least likely to buy option-free bond (falls most)

Key rate durations ★★★★★

In plain words: a key rate duration shows the price change if only one point on the curve moves. They add up to effective duration and reveal slope and curvature bets.

- Zero-coupon bond: KRD = its modified duration × weight
- Active KRD > 0 at a maturity → expects that rate to fall; < 0 → expects it to rise
- Example: active duration −3.875 (bearish), +0.98 short end, 0 long end, large negative middle → **bear steepener + negative butterfly**
- KRDs +3, −0.5, −3 at 10/20/30-yr via derivatives → receive-fixed 10-yr, short 20-yr futures, pay-fixed 30-yr

Across currencies: unhedged, hedged, carry ★★★★★

$$R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1$$ (R_FX from the DIRECT quote, DC/FC)

Australian buys US 2-yr zero at 96.374 → 99.939 after 1 yr: $R_{FC} = 3.70\%$

USD/AUD 0.90 → 0.80 ⇒ AUD/USD 1.1111 → 1.25: $R_{FX} = 12.5\%$ → $R_{DC} = 1.037 \times 1.125 - 1 = \mathbf{16.66\%}$

CIRP: $F = S \times [(1 + r_{DC})/(1 + r_{FC})]^T$

US yield $(100/96.374)^{½} - 1 = 1.864\%$; AUD 5% → $F = 1.1111 \times (1.05/1.01864)^2 = \mathbf{1.1806}$

Hedged: 100 × 1.1806 = 118.06 vs cost 107.08 → $(118.06/107.08)^{½} - 1 = \mathbf{5.0\%}$ = domestic rate

- CIRP: high-rate currency at a forward **discount** → hedged foreign bond earns the **domestic** rate
- UIRP says high-rate currencies depreciate — usually fails → **forward rate bias** → carry trades profit on average
- Hedge a coupon bond with a **fixed-fixed cross-currency swap** (pay foreign, receive domestic) = 2 fixed-for-floating swaps + cross-currency basis swap (exchanges principal); dropping one leg leaves a rate view with currency still hedged
- Negative basis: non-USD leg below CIRP fair rate (borrowers of USD pay up)

Carry: borrow USD 1%, roll 90-day INR at 5% → $(1.0125)^4 - 1 = 5.1\%$ → +4.1% if stable

INR rates 5,6,4,3% → 4.6%; INR/USD 70 → 85: USD/INR 0.014286 → 0.011765 = -17.6%

Total = 4.6 - 1.0 - 17.6 = **-14.0%** (flight to quality crash risk)

- Cross-market swaps: receive fixed where rates will fall, pay fixed where they will rise; receive floating where rates will rise
- Foreign bull steepener + stable domestic → pay foreign floating, receive domestic fixed

TRAP Excess carry returns mean **UNCOVERED** parity fails — covered parity is not involved.

Evaluating a yield curve strategy (worked) ★★★★★

CAD zero-coupon, US investor, CAD -0.5%, yields +25 bp

Bullet: rolldown $(98.75 - 96.75)/96.75 = 2.067\%$; yield $-4.89 \times 0.0025 + ½ \times 28.5 \times 0.0025^2 = -1.214\%$

Barbell: rolldown $(98.95 - 97)/97 = 2.010\%$; yield $-4.92 \times 0.0025 + ½ \times 43 \times 0.0025^2 = -1.217\%$

USD: bullet $(1 + 0.02067 - 0.01214)(0.995) - 1 = \mathbf{0.35\%}$; barbell **0.29%**

Bullet wins on rolldown (sits on a steeper part of the curve)

Quiz: barbell 60% 2-yr + 40% 10-yr, bear flattener 2-yr +50, 10-yr +30

$0.6 \times (-0.92\%) + 0.4 \times (-2.81\%) = \mathbf{-1.68\%}$

- Schweser recommends compounding currency (multiplicative); adding is unlikely to be marked wrong

06 P6 Fixed-Income Active Management: Credit Strategies

≈19% of PM marks · 12 subtopics · chapter priority ★★★★★

Credit risk, spreads and the credit cycle ★★★★★

In plain words: a credit spread is the extra yield a risky bond pays over a safe government bond. It pays you for two risks: the issuer may not pay (**credit risk**) and you may not be able to sell easily (**liquidity risk**).

```
Expected loss = POD × LGD ; LGD = 1 - recovery rate
CVA = PV of Σ(POD × LGD × expected exposure) → risky bond
= risk-free bond - CVA
Simple fair spread ≈ POD × LGD
First lien spread 1.5%, recovery 50% → POD = 1.5%/0.5 =
3%
Second lien recovery 40% → spread = 3% × 0.6 = 1.8%
```

IG = BBB and above (main risk = **credit migration**/downgrade); HY = BB and below (main risk = default)

Credit spread curve = spread vs maturity; upward = defaults more likely later

Cycle: peak → spreads rising, curve upward sloping; recession → IG flattens, **HY inverts** (near-term default risk); HY moves more sharply

Rates and spreads move oppositely → HY bond price reacts less to rate changes: **empirical duration < analytical duration**

```
HY 6% vs Treasury 2% (spread 4%), MD 4.4: Treasury falls
to 1%, spread widens to 5% → yield still 6% → price
unchanged (empirical duration ≈ 0)
```

Fixed-rate spread measures ★★★★★

Measure	Definition	Note
Yield (benchmark) spread	Bond YTM − nearest on-the-run govt YTM	Simplest; maturity mismatch
G-spread	Bond YTM − **interpolated** govt YTM	Upward curve → g-spread < yield spread
I-spread	Bond YTM − interpolated **swap** rate	Tradeable hedge; carry
Asset swap spread (ASW)	Bond **coupon** − swap rate	Spread over MRR via bond + pay-fixed swap; bond below par → i-spread > ASW
Z-spread	Constant spread over the govt **spot** curve that prices the bond	Respects term structure
CDS basis	CDS spread − z-spread	Technical differences
OAS	Spread over an interest-rate tree after removing the option	**Only** valid measure for bonds with options — best for comparisons

```
Interpolation weight on longer bond = (M - S)/(L - S)
12-yr corporate 4.5% (coupon 4.3%, MD 11.1); govt 10-yr
3.0%, 20-yr 5.0%
Yield spread = 4.5 - 3.0 = 1.5% ; weight on 20-yr = 2/10
= 0.2 → interpolated 3.4% → g-spread = 1.1%
20-yr falls 10 bp → interpolated 3.38% → corporate yield
4.48% → price +11.1 × 0.02% = +0.222%
Swap rates 3.2% / 5.3% → interpolated 3.62% → i-spread =
0.88%, ASW = 4.30 - 3.62 = 0.68%
```

TRAP A bond whose OAS is very different from its other spreads has an embedded option — compare it using **OAS** .

Floating-rate note spreads ★★★★☆

```
Coupon = MRR + quoted margin (QM) ; required margin =
discount margin (DM)
DM > QM → below par ; DM < QM → above par ; DM = QM → par
1-yr FRN: MRR 0.55%, QM 1.25%, DM 1.55% → below par
Quarterly coupon (0.55 + 1.25)/4 = 0.45% → $4,500 ;
discount (0.55 + 1.55)/4 = 0.525%
Value of $1m ≈ $997,040
```

FRNs: low rate duration, but spread duration ≈ like a fixed bond (QM fixed)

DM assumes flat MRR; **Z-DM** uses forward MRR + spot discounting; upward MRR curve (recovery) → Z-DM < DM

Spread duration, DTS and excess spread ★★★★★

```
Spread return ≈ -EffSpreadDur × ΔSpread + ½ ×
EffSpreadCon × ΔSpread²   (Δspread = ΔOAS)
DTS = EffSpreadDur × spread
Portfolio: 70% A bond (9.0 × 180 = 1,620) + 30% C bond
(9.5 × 400 = 3,800) → DTS = 2,274
Index OAS 200 → +20 bp = +10% → portfolio falls 2,274 ×
0.1 = 227.4 bp
Expected excess spread = spread × t - EffSpreadDur ×
ΔSpread - POD × LGD × t
Spread 250 bp, dur 4.5, POD 1.5%, LGD 55%, spread → 270
in 6 months:
0.0125 - 4.5 × 0.0020 - 0.015 × 0.55 × 0.5 = -0.0625% →
overvalued
```

Spread changes are proportional to spread level → use DTS

Portfolio OAS is market-value weighted (include accrued interest)

TRAP De-annualise spread and POD × LGD for the holding period, and work in decimals.

Bottom-up credit: models & relative value ★★★★★

Model	How	Examples
Structural	Default when asset value falls below liabilities ('distance to default'); asset volatility from equity data	Moody's EDF, Bloomberg DRSK
Reduced form	Default intensity from macro + firm ratios	**Altman z-score**

```
z = 1.2 × WC/TA + 1.4 × RE/TA + 3.3 × EBIT/TA + 0.6 ×
MVE/TL + 1.0 × Sales/TA
> 3 low risk · 1.8–3 some risk · < 1.8 default likely
Example: 0.6, 0.54, 0.17, 2.5, 0.57 → z = 4.11 → low risk
```

```
New 10-yr issue: 5-yr spread 1.5 - 0.8 = 0.7% ; 15-yr
govt interpolated 0.75×2% + 0.25×4% = 2.5% → 15-yr spread
1.0%
10-yr spread = 0.5×0.7 + 0.5×1.0 + 0.05 premium = 0.90%
```

Bottom-up: universe → sectors → best relative value; operating history, profitability, leverage, coverage ratios

Lower equity volatility → farther from default → lower POD

Watch structure (options, seniority), liquidity, M&A

Top-down, factor and ESG credit strategies ★★★★☆

- Macro: strong growth → spreads narrow → move **down** in quality (more HY); weak growth → move up
- High spreads alone are not an opportunity — only if you foresee improvement before others
- Average rating: linear scale (1,2,3…) **understates** credit risk; non-linear (Moody's 1, 10, 20, 40…) is better; use DTS
- Credit factors: **carry** (OAS), **defensive** (low-risk/short duration earns better risk-adjusted), **momentum**, **value**
- ESG: negative screening · ESG ratings (correlated with credit ratings) · direct funding e.g. **green bonds**

Liquidity risk & tail risk (VaR) ★★★★★

On-the-run sovereigns very liquid; small off-the-run corporates may not trade → matrix pricing; TRACE for US corporate trade data

Manage liquidity: liquid bonds for short-term strategies · CDS and ETFs (APs create/redeem) · **pay-fixed asset swap** while slowly selling a big illiquid bond, unwinding as you sell

VaR method	Pro	Con
Parametric (normal; 5% = 1.65σ, 1% = 2.33σ)	Simple	Poor for non-normal (options)
Historical simulation	Real data, non-normal OK	Depends on the sample
Monte Carlo	Flexible	Model risk

```
$100m bond, MD 12, annual yield vol 1.5% → monthly 1.5%/√12 = 0.433%
1% VaR yield move = 2.33 × 0.433% = 100.89 bp → loss 12 × 1.0089% = 12.107% → $12.1m
```

VaR flaws: tails worse than forecast, ignores correlation/liquidity shifts, no tail-size info

CVaR = average loss in the tail · **IVaR** = VaR change from adding/removing a position · **Relative VaR** vs benchmark

Stress the consequences: margin calls, sponsor contributions, bank reserves

Credit default swaps: pricing ★★★★★

In plain words: a CDS is insurance on a bond. The **buyer** of protection pays a fee and gets paid if the issuer defaults — he is 'short risk', like being short the bond. The **seller** is 'long risk', like owning the bond.

```
Standard coupon: 1% IG, 5% HY
Upfront premium (paid by buyer) ≈ (CDS spread - fixed coupon) × EffSpreadDur
CDS price ≈ 1 + (fixed coupon - CDS spread) × EffSpreadDur
HY, spread 5.5%, dur 4.5 → upfront 2.25% ; price 97.75
Spread falls to 5.4% → +0.45% price → buyer loses 0.45%
Quiz: HY 4.7%, dur 4.5, $10m → seller pays buyer (0.05 - 0.047) × 4.5 × 10m = $135,000
```

Single-name or index (CDX NA IG = 125 names); payer CDS option = right to buy protection (profits if spreads above strike); receiver = right to sell protection

Long risk position behaves like a long bond: profits when spreads fall and CDS price rises

CDS long/short and curve trades ★★★★★

```
Long XYZ / short ABC: sell XYZ protection, buy ABC protection, $15m each
ABC +15 bp: 15m × 0.0015 × 4.59 = $103,275 ; XYZ -20 bp: 15m × 0.002 × 4.62 = $138,600 → $241,875
Flattener (HY index): buy 5-yr protection $5m (dur 4.78 → BPV $2,390), sell 10-yr protection $2,390/(9.18×0.0001) = $2,603,486
5-yr -5 bp → -$11,950 ; 10-yr -20 bp → +$47,800 → +$35,850
```

Buy protection where spreads will widen; sell where they will narrow

Curve trade: buy protection at maturities expected to rise relative to others

Static & dynamic credit curve strategies ★★★★★

View	Cash bonds	CDS
Static curve	Lower average rating or extend spread duration (buy and hold longer)	Sell protection on lower quality or longer tenors
Economic slowdown (HY widens more; curves flatten/invert)	Buy IG, sell HY (flight to quality)	Buy HY protection, sell IG; buy short-dated, sell long-dated protection
Economic recovery (HY narrows more; curves steepen)	Buy HY, sell IG	Sell HY protection, buy IG; sell short-dated, buy long-dated protection

```
Static, cash: overweight $10m 15-yr corporate vs 10-yr for 6 months
Extra coupon $35,000 (govt part $15,000, credit $20,000)
Rolldown: 10-yr → 100.8888 to 101.1053 = $21,650 ; 15-yr → 102.4187 to 103.0726 = $65,390 → extra $43,740
Total $78,740; from credit = $20,000 + $32,695 = $52,695
```

```
Static, CDS: sell $50m 10-yr HY protection; 5-yr 450 bp, 10-yr 550 bp → 9-yr 530 bp
Price 0.9555 → 1 + (0.05 - 0.053) × 8.1 = 0.9757 → +$1,010,000 + coupon $2,500,000 = $3.51m
Slowdown: sell IG protection + buy HY protection, $50m each, spreads +40% / +80%
Net coupon -$2m ; HY leg +$4,832,500 ; IG leg -$770,000 → +$2,062,500
```

```
Quiz: sell IG protection $10m, spread 150 → 120 bp, dur 4.5 → 3.5
Price 0.9775 → 0.993 = $155,000 + coupon $100,000 = $255,000
```

Global credit, emerging markets ★★★★☆

One-industry/commodity EM economies give little sector diversification

Sector mix differs (US has far more MBS/ABS)

Accounting differences distort ratios

Credit cycles differ in timing → opportunities

```
Recovery view, buy European HY credit via CDS (sell protection): spread 4.5%, dur 4.55
Price 1 + (0.05 - 0.045) × 4.55 = 1.02275 → spreads -20% → 1 + (0.05 - 0.036) × 4.55 = 1.0637 → +4%
EUR +2% → 1.04 × 1.02 - 1 = 6.08%
```

Sovereign analysis: institutions (property rights, politics, geopolitics, ESG) · economy (tax base, debt/GDP, deficit/GDP — fiscal stability) · FX regime (external debt/GDP = distress risk; reserves/GDP = liquidity)

EM corporates: government influence, **sovereign ceiling**, liquidity premium, currency risk (UIRP often fails)

Structured credit & fixed-income analytics ★★★★☆

Reason to use	Instruments
Tranching creates new risk profiles (losses hit junior first)	CDOs (general debt), CLOs (leveraged loans)
Access collateral you can't buy directly	MBS, ABS (cards, autos), **covered bonds** (on bank's balance sheet, recourse to the bank, bad loans substituted → lower risk)

Recovery + housing rebound → sell AA MBS tranche, buy BB MBS tranche

Analytics = inputs (positions, market data, credit/ESG ratings, index data) + user parameters (term-structure, VaR, scenario models, filters, horizon, objective) + outputs (exposures vs benchmark, construction, risk/scenario, trading & cash tools)

MBS hold a short prepayment option → **lower** volatility assumption raises their value

07 P7 Trade Strategy and Execution

≈12% of PM marks · 8 subtopics · chapter priority ★★★★☆

Why managers trade ★★★★★

Motivation	Key points
Profit seeking	**Alpha decay** → urgency (news traders high, fundamental low); avoid **information leakage** using several venues incl. **dark pools** (ATS in US; MTF / systematic internaliser in Europe) vs **lit** exchanges
Risk management / hedging	Rebalance to target exposure (duration, FX hedge); derivatives if liquid and allowed; watch leverage
Cash flows	Subscriptions/redemptions; **cash drag** → equitise with ETFs/futures; redemptions at NAV → trade at **close**; weigh liquidity and tax when choosing what to sell
Corporate actions, margin calls, index reconstitution	Reinvest income; urgent sales for margin calls; reconstitution → trade at close to cut tracking error

Inputs to trading strategy & the trader's dilemma ★★★★★

Input	Effect
Order: side, absolute size, size vs ADV	Buying into rising market costlier; one-sided lists carry more market risk; bigger orders → more impact
Security: type, short-term alpha, volatility, liquidity	Fast alpha decay → urgent; high volatility → **execution risk**; liquidity (tight spreads, depth) lowers both costs
Market conditions	Crises: volatility up, liquidity down (2008; quant crowding Aug 2007)
Risk aversion	More averse → trades faster

Trader's dilemma: trade fast → market impact & leakage; trade slow → execution risk

Longer horizon lowers market impact but raises execution risk

Higher liquidity lowers **both**

Reference price benchmarks ★★★★★

Type	Benchmark	Used by
Pre-trade	Decision price	Ideal measure of the PM's decision
	Previous close	Quant managers (proxy for decision)
	Opening price	Fundamental long-term managers (ignores overnight news); not if trading in the opening auction
	Arrival price (when order reaches market)	Alpha-seeking active managers
Intraday	VWAP	Passive over the day, rebalancing (sells fund buys)
	TWAP	Ignore volume outliers; erratic volume
Post-trade	Closing price	Index funds, NAV-based redemptions; can't judge during trading
Price target	Manager's fair value	Short-term alpha: buy all you can below target

Short-term alpha trade: high urgency, price target + arrival price, algorithm

Long-term alpha in illiquid bonds: low urgency, sell gradually over weeks

Index reconstitution, low TE mandate → **market on close**

Execution methods & algorithms ★★★★★

Method	Description
High-touch principal (broker risk)	Dealer takes risk, priced into spread; OTC, RFQ markets; very large/illiquid/urgent
High-touch agency	Broker finds the other side; risk stays with manager
Electronic / DMA	Order-driven limit order book; buy side accesses exchange through broker's technology
Profit-seeking algos	Market makers, quants, HFT

Execution algorithm	Best for
Scheduled: POV (participation), VWAP, TWAP	Small orders, liquid markets, low urgency, minimise impact (rebalancing)
Liquidity-seeking (opportunistic)	Large orders, illiquid, higher urgency, fear of leakage, sporadic liquidity; lit + dark
Arrival price	Small orders, liquid; prices expected to move against you; risk-averse to execution risk
Dark strategies / liquidity aggregators	Large orders, illiquid, need not fill immediately
Smart order router (SOR)	Small market or limit orders across many venues

POV: uses liquidity when it appears but trades at any price and may not finish

VWAP/TWAP: finish on time but may force trades in thin periods

100% of ADV → high-touch principal; big + urgent + illiquid → liquidity-seeking; small, low urgency → scheduled

New: **clustering** (ML groups trades to pick algos) · high-frequency forecasting with **LASSO** to cut variables

Trading by market ★★★★☆

Market	How it trades
Equities	Lit exchanges + dark pools; electronic; algos common
Fixed income	Many heterogeneous issues, low liquidity, big tickets → dealer quote-driven, high-touch, electronic **RFQ**; algos only for on-the-run Treasuries & futures; urgent → principal, else agency
Exchange-traded derivatives	Electronic, DMA; algos less common (more futures than options)
OTC derivatives	Dealer, high-touch; post-2008 central clearing push (Dodd-Frank 2010)
Spot FX	OTC tiers: interbank, interdealer, bank-to-client; large urgent → RFQ; large non-urgent → scheduled algos or agency; small → DMA

Example: small urgent futures → DMA · agency MBS → high-touch broker · large non-urgent equity → scheduled algorithm

Implementation shortfall (worked) ★★★★★

```
IS = paper return - actual return  (paper = all shares at
decision price, no costs)
Buy 50,000 SJB, decision $20.00, arrival $20.10, filled
40,000 at $20.34, commission $0.02, close $20.55
Paper = 50,000 × 0.55 = 27,500 ; actual = 40,000 × 0.21 -
800 = 7,600 → IS = $19,900 = 199 bp
Delay = 40,000 × (20.10 - 20.00) = 4,000 (40 bp)
Trading = 40,000 × (20.34 - 20.10) = 9,600 (96 bp)
Opportunity = 10,000 × (20.55 - 20.00) = 5,500 (55 bp)
Fixed fees = 800 (8 bp)
```

```
Sell 100,000 FTRB, decision £2.56, arrival £2.59, sold
70,000 at £2.60, fee £400, close £2.54
Paper = 100,000 × 0.02 = 2,000 ; actual = 70,000 × 0.06 -
400 = 3,800 → IS = -£1,800 = -70.3 bp (benefit)
Delay -82.0 · trading -27.3 · opportunity +23.4 · fees
+15.6
Quiz: sell 10,000 at $32.50, 4,000 filled at $32.65,
$0.03/sh, close $32.35 → paper 1,500, actual 1,080 →
12.92 bp
```

Delay cost ↓ with faster pre-trade analysis

Opportunity-cost analysis → redeploy unfilled cash into the next idea

Negative cost = trader added value

TRAP For sells, flip the signs: a higher sale price is a gain.

Evaluating execution: arrival, market-adjusted, added value ★★★★★

```
Cost (bp) = Side × (execution price - benchmark) /
benchmark × 10,000   (Side +1 buy, -1 sell)
Index cost (bp) = Side × (index VWAP - index arrival) /
index arrival
Market-adjusted cost = arrival cost - β × index cost
Added value = arrival cost - estimated pre-trade cost
Buy at $25.50 vs close $25.60 → -39.1 bp (benefit)
Buy: arrival €10 → €10.15 = 150 bp ; index 3,500 → 3,507
= 20 bp ; β 1.5 → 150 - 30 = 120 bp
Pre-trade estimate 160 → added value -10 bp (good)
Quiz sell: 8.40 → 8.38 = 23.8 bp ; index cost -1 ×
50/3000 = -166.7 ; β 1.3 → 240.5 bp
```

Market-adjusted cost stops traders being blamed or credited for market moves

Trade governance & best execution ★★★★★

Trade policy area	Contents
1. Meaning of best execution	Best **result**, not just best price: price, costs, speed, likelihood of execution & settlement, size, nature of trade (block sale at a discount can be best execution)
2. Factors for optimal approach	Urgency, size, liquidity, security type, venues, strategy horizon, reason; regulations — **MiFID II** effectively bans soft dollars in Europe
3. Approved brokers & venues	Kept by a **Best Execution Monitoring Committee** (execution, compliance, risk); criteria: execution quality, financial stability, ethics, settlement, costs, willingness to commit capital
4. Monitoring	Reputation, errors, crimes, solvency; remove failing brokers; keep records for clients and regulators

TRAP 'Use the lowest-cost broker' and 'traders alone decide governance' are both **NOT** best practice.

08 P8 Case Study: Institutional (University Endowment)

≈15% of PM marks · 10 subtopics · chapter priority ★★★★★

How to read this case ★★★★★

In plain words: this reading has almost no new theory. It takes one university endowment (QU) and asks you to apply ideas from the whole curriculum: liquidity, asset allocation, ethics, derivatives, rebalancing and ESG. Expect it as an **essay set**: 'justify', 'recommend', 'identify the violation'.

Answer with the case numbers: quote the % or bp from the exhibit, then give the reason

One recommendation + one reason per point earns the marks; long lists do not

Managing liquidity risk: four tools ★★★★★

Tool	What it does
1. Liquidity profile & time-to-cash table	Sorts assets by how fast they turn into cash (e.g. highly liquid, liquid, semi-liquid, illiquid). **Liquidity budget**: minimums for liquid tiers, a **maximum** for the illiquid tier. Exit > 1 year = illiquid. Same asset can differ by vehicle (ETF vs commingled fund)
2. Rebalancing & commitment policy	**Systematic rebalancing**: calendar or percent-range (wider ranges for volatile assets). **Automatic adjustment**: PE 2% under target (β ≈ 1) → hold +2% public equity → same systematic risk, less illiquidity. **Commitment pacing** model (drawdown and distribution rates); diversify across funds and vintages; scenario analysis
3. Stress testing	Test assets **and** liabilities together: high volatility, low returns, high correlation, low liquidity, higher payouts. Base on history, statistical models or hypothetical scenarios
4. Derivatives	Little cash needed. Futures overlay rebalances without touching external managers. Buy options = cheap leverage; sell options = premium income (liquidity)

TRAP Calendar and percent-range rules are **SYSTEMATIC** rebalancing policies — a quiz statement that calls them something else is wrong.

The illiquidity premium ★★★★☆

In plain words: investors want extra return for money they cannot get back quickly. The longer the lock-up, the bigger the premium.

```
Option view: price of illiquid asset = price of same
marketable asset - value of a put (the right to sell you
gave up)
Illiquidity premium = expected return of illiquid asset -
expected return of marketable asset
```

Hard to measure: many other factors drive returns

Estimates use broad indexes, but a real investor holds few funds → more **unsystematic** risk → needs a liquidity budget

TRAP 'The illiquidity premium is easy to measure' is FALSE .

QU endowment: facts you need ★★★★★

Item	Detail
Size	$8bn: 75% unrestricted, 25% donor-restricted
Role	Pays about 60% of the $583m operating budget
Spending rule	Changed from 5% of beginning value to a **geometric smoothing** rule (uses prior spending $358.1m and prior year-end value $7,002.3m)
Return need	7–8% nominal = 5% spending + 2–3% inflation
Risk target	Standard deviation 12–14%
Manager	QUINCO: 11-person board, 13 staff, **external managers only**, 5 asset classes; Board approves managers; CIO Winter leads the investment committee

Asset class	1996	2017
Cash	1%	1%
Domestic stocks + bonds	~70%	~30%
International developed equity	25%	10%
Emerging markets	0	10–15%
Private equity	< 5%	15–20%
Real assets	< 5%	10–15%
Diversifying strategies	0	10–15%

Review: PE returns are smoothed → **unsmooth** them and volatility rises

QU lagged peers because it took less risk and held fewer illiquid assets

Proposed allocation: justify it & name the trade-offs ★★★★★

In plain words: CIO Thompson wants more private equity and real assets. You must argue for it, name what it costs, and say what to check.

Metric	Existing	Proposed
Sharpe ratio	0.33	0.34
Standard deviation	12.5%	13.2% (+0.7%)
P(keep ≥ 75% purchasing power, 20 yrs)	65%	70%

Why more illiquids: long horizon lets QU earn the illiquidity premium → efficient frontier shifts up · 20 years of good private-asset results with expert managers · QU is under-weight illiquids vs peers

Thompson's reasons: lower expected returns everywhere · QU takes less risk than peers · fixed income has a low Sharpe ratio · Monte Carlo shows better purchasing power

Trade-offs: higher volatility (but needed to earn the same return) · higher fees (exhibit returns are already **net of fees**)

Reduce the new risk: many smaller commitments (less unsystematic risk) or cash-flow modelling under stress

Before approving, confirm: 13.2% SD and 30% chance of 25% erosion fit risk appetite · new mix still meets all liquidity needs

TRAP More illiquids raise UNSYSTEMATIC risk — diversify across many funds and vintages.

Why liquidity gets worse in a crisis ★★★★★

Cause	Effect
Capital calls > distributions	PE funds call cash (a legal duty) while exits stop
Gates / lock-ups	Hedge funds may block withdrawals
Lagged PE valuations (denominator effect)	Public assets fall now, PE marks fall later → PE share rises, liquid share falls
Spending rule is countercyclical	Smoothed spending is > 5% of value after a fall → more cash needed exactly when markets are weak

Profile (highly liquid / liquid / semi / illiquid)	Normal	Stress
Existing	19 / 26 / 22 / 33	15 / 26 / 20 / 39
Proposed	14 / 24 / 23 / 39	11 / 25 / 21 / 43

Proposed: highly liquid −5% and illiquid +6% (normal); −4% and +4% (stress)

Follow-up: cash-flow forecasts, commitment pacing, liquidity budget, regular stress tests (historical + hypothetical), monitoring at high-risk times

Ethics in manager selection (who broke what) ★★★★★

In plain words: a donor-trustee's firm (GVC) is a finalist for a QU mandate. Many people act badly. For each person, name the Standard and the fact.

Person	Standards	Why
Aaron Winter, CIO	VI(A), I(B), III(E), I(C)/III(D), V(A)	Did not disclose GVC owner's link to QU or that GVC's MD is a former colleague; under pressure; suspected confidential data and inflated returns but did not report or investigate
Virginia Hall, trustee & GVC owner	VI(A), IV(A), III(E)	Conflict (trustee + finalist owner); put own business before QU; may have obtained confidential data
QU president & treasurer	IV(A), III(E), VI(A)/I(B)	Pressured Winter to hire GVC; may have leaked data; should disclose bias and abstain from the vote
Jason Allen, GVC MD	I(C), III(D)	Presented possibly wrong / inflated performance
Bud Davis, senior PM	VI(A), I(B)	Spouse is Hall's daughter; recommended GVC without disclosing

Standards list: I(B) Independence & Objectivity · I(C) Misrepresentation · III(D) Performance Presentation · III(E) Confidentiality · IV(A) Loyalty · V(A) Diligence · VI(A) Disclosure of Conflicts

GVC context: raising $300m Fund II after a $100m Fund I; it cut its fee; rival finalist Beacher

TRAP Non-CFA people (president, treasurer) are judged by the same tests in this case answer — apply the Standards to each actor named.

Equity exposure: ETF vs futures vs total return swap ★★★★★

In plain words: QU wants +1% US large-cap ($80m) for tactical asset allocation. Compare cost, work, tracking error and risk.

	ETF	Futures	Total return swap
Cost	**35 bp** (70% impact + fee)	42 bp (half is roll cost)	51 bp
Logistics	None (ETF manager runs it)	Quarterly roll, daily margin	None
Tracking error	Cash drag, diversification rules	Liquidity, rate differences	**None** (receives index return)
Rate / counterparty risk	None	Interest rate risk	Rate + counterparty
Overall (no leverage)	**Best**		

```
With leverage 4 (25% cash, 75% financed): financing 2%
for futures/swaps, 2.5% for ETF
ETF: 80m × 75% × 2.5% = $1.5m = 1.875% → total 0.35 +
1.875 = 2.225%
Futures: 80m × 75% × 2% = $1.2m = 1.5% → total 0.42 + 1.5
= 1.92% (cheapest by 9 bp)
Swap: 1.5% + 0.51 = 2.01%
```

Policy changes: tracking error limit 100 → 250 bp (150 bp for TAA); max leverage 5%; TAA uses fair-value / mean-reversion models (1–3 yrs, backtested)

ETF on 50% margin: $80m exposure = $40m cash + $40m borrowed

Rebalancing: cash market or derivatives? ★★★★★

Policy: rebalance quarterly; check drift monthly; liquid classes back to target or corridor edge; illiquid classes via new commitments

Now: int'l developed equity 11.5% (top of range 11%) · fixed income 6.5% (target 9%, range 6–12%) · PE 19.2% (low end 18%) · real assets 13.8% (low end 13%)

Trade	Cash market	Futures	Choose
0.5%, ~3 months	30 bp (spread 5 + impact 5 + cash drag 20)	24 bp (3 + 4 + tracking 17)	**Futures**
2.5% (to 9% target), ~1 year	60 bp (5 + 5 + cash drag 50)	82 bp (4 + 4 + tracking 68 + roll 6)	**Cash** (saves 22 bp)

Cash market: cash drag (delays, disrupting active managers) = two-thirds of cost

Futures: short equity + long bond futures → two contracts → tracking error 17 bp vs 8 bp for one; tracking error > 70% of cost

Rule: small, short-term, reversible, needs speed → derivatives overlay (no effect on active managers). Large, long-term, permanent → cash market

TRAP Derivatives' three roles here: **REBALANCING, TAA CHANGES, SHORT-TERM LIQUIDITY** .

ESG policy & the child-labour case ★★★★★

In plain words: a portfolio company (Portro) used a supplier with child labour (ILO report). Students demand divestment. What should QU do?

QUINCO ESG policy: ESG and financial factors both affect risk and return · transparency · long-term sustainability

Manager due diligence (added by Natalya Long, CFA): formal ESG policy? · how ESG enters the process · timely disclosure of material ESG issues · fit with QUINCO policy

Responsible ownership: proxy voting and engagement (human capital, climate, disclosure) · reporting · communication · UN PRI signatory

Monitoring metrics: ESG scores vs peers and policy benchmark · sensitivity to climate risks · portfolio carbon footprint · track trends over time

Response: engage the manager and Portro · join other investors · vote proxies · remove the supplier's goods from the campus store · divest only as a **last resort**

TRAP Divestment is **NOT** the first answer — engagement is preferred.

F Formula Sheet (all readings)

≈revise last marks · 4 subtopics · chapter priority ★★★★★

Equity (P1–P3) ★★★★★

```
Tracking error = SD(Rp - Rβ)
E(R_A) = IC × √BR × σ_A × TC
Active Share = ½ Σ |wpi - wβi|
Contribution to variance CVi = wi × Cov(ri, rp) ; % = CVi / σp²
Contribution to active variance = (wpi - wβi) × Σj (wpj - wβj) Cov(RAi, RAj)
Gross exposure = long + |short| ; net = long - short
R_g ≈ R_a - σ²/2
```

Liability-driven & yield curve (P4–P5) ★★★★★

```
MD = Macaulay duration / (1 + periodic YTM)
%ΔValue ≈ -MD × ΔY + ½ C (ΔY)²
BPV = MD × value × 0.0001
Futures BPV ≈ BPV_CTD / CF_CTD ; contracts = (BPV_L - BPV_A) / BPV_futures
Swap NP = BPV gap / swap BPV per 1 of notional ; swap BPV = (D_fixed - D_float) × 0.0001
Effective duration = (PV- - PV+) / (2 × Δcurve × PV0)
Effective convexity = (PV- + PV+ - 2PV0) / (Δcurve² × PV0)
Key rate duration = -(ΔPV / PV) / Δkey rate
Butterfly spread = -short + 2 × medium - long
R_DC = (1 + R_FC)(1 + R_FX) - 1
Forward (CIRP) F = S × [(1 + r_DC)/(1 + r_FC)]^T
```

Bonds with options: use effective, not modified, duration

Credit (P6) ★★★★★

```
Credit spread ≈ POD × LGD
EffSpreadDur = (PV- - PV+) / (2 × Δspread × PV0)
EffSpreadCon = (PV- + PV+ - 2PV0) / (Δspread² × PV0)
%Δprice ≈ -EffSpreadDur × Δs + ½ × EffSpreadCon × (Δs)²
%Δrelative value ≈ -spread duration × Δspread
DTS = EffSpreadDur × spread
Expected excess spread ≈ spread - (EffSpreadDur × Δspread) - (POD × LGD)
FRN discount margin: price from (MRR + DM) vs coupon MRR + QM
CDS price ≈ (fixed coupon - CDS spread) × EffSpreadDur_CDS
```

Trading (P7) ★★★★★

```
IS = paper return - actual return
= delay + trading (execution) + opportunity cost + fixed fees
Cost (bp) = Side × (P_exec - P_bench)/P_bench × 10,000 (buy +1, sell -1)
Market-adjusted cost = arrival cost - β × index cost
Added value = arrival cost - pre-trade estimated cost (negative = trader added value)
```